

Assessment week 3 | Case Study 1

Smart City Traffic & Accident Analytics

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(Batch 3)



https://github.com/amankumar1ext-web/Case_Study/tree/main/W3_Case_Study_1

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CHAPTER 1: INTRODUCTION

1.1 Project Overview

The project Smart City Traffic & Accident Analytics focuses on analyzing traffic and accident data to support better traffic management and road safety. The project uses Python, MySQL, SQL, and Power BI to process data, perform analysis, and present meaningful insights through an interactive dashboard.

1.2 Problem Statement

MetroCity is experiencing increasing traffic congestion and frequent road accidents. Although traffic sensors and surveillance systems are available, the administration needs a data-driven approach to analyze traffic patterns, identify accident-prone locations, and understand congestion during peak hours.

1.3 Business Objectives

The main objectives of the project are:

- Analyze traffic patterns across different locations.
- Identify accident-prone areas using historical accident data.
- Analyze congestion hotspots during peak hours.
- Analyze traffic flow, congestion levels, and average speeds.
- Generate insights to support traffic management and road safety.

1.4 Project Scope

The project covers the integration and analysis of traffic sensor, traffic, and accident data. Python is used for data processing and analysis, MySQL is used for structured data storage and SQL analysis, and Power BI is used to create interactive visualizations and dashboards.

1.5 Implementation Approach

The project follows the workflow:

Data → Python/Pandas → Data Validation & Cleaning → MySQL → SQL Analysis → Power BI → DAX → Dashboard → Business Insights

CHAPTER 2: DATA LOADING AND VALIDATION

2.1 Load the datasets

Load the datasets and understand. There are two datasets for traffic and accident analytics. the road traffic sensor dataset and the traffic accident dataset.

Python and Pandas were used to load the CSV datasets into Jupyter Notebook for initial inspection and validation.

2.2 Data Validation

The datasets were validated by checking their dimensions, column names, data types, missing values, duplicate records, and unique identifiers.

```
SENSOR DATA SHAPE:
(300, 6)

ACCIDENT DATA SHAPE:
(5000, 10)
```

```
SENSOR DATA MISSING VALUES:
Sensor_ID      0
Location       0
Date_Time      0
Vehicle_Count  0
Average_Speed  0
Congestion_Level 0
dtype: int64
```

```
SENSOR DATA INFO:
<class 'pandas.DataFrame'>
RangeIndex: 300 entries, 0 to 299
Data columns (total 6 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Sensor_ID           300 non-null   str
1   Location            300 non-null   str
2   Date_Time           300 non-null   str
3   Vehicle_Count       300 non-null   int64
4   Average_Speed       300 non-null   int64
5   Congestion_Level    300 non-null   str
dtypes: int64(2), str(4)
memory usage: 14.2 KB

ACCIDENT DATA INFO:
<class 'pandas.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 10 columns):
#   Column              Non-Null Count  Dtype
---  ---
0   Accident_ID         5000 non-null   str
1   Date_Time           5000 non-null   str
2   Location            5000 non-null   str
3   Weather_Condition   5000 non-null   str
4   Road_Condition      5000 non-null   str
5   Vehicle_Type        5000 non-null   str
6   Accident_Severity   5000 non-null   str
7   Number_of_Vehicles  5000 non-null   int64
8   Casualties          5000 non-null   int64
9   Traffic_Density     5000 non-null   str
dtypes: int64(2), str(8)
memory usage: 390.8 KB
```

After observing and validating Dataset it is clean and organised. No need to clean or organise as there is no duplication, Null or missing value.

Dataset	Rows	Columns	Missing Values	Duplicates
Sensor Data	300	6	0	0
Accident Data	5,000	10	0	0

Data Observation:

- Sensor records: 300
- Unique Sensor IDs: 300
- Accident records: 5,000
- Unique Accident IDs: 5,000
- Missing values: None
- Duplicate records: None
- Sensor locations: 5
- Accident locations: 5

Both datasets contain the same 5 locations.

Date Coverage:

- Sensor data: 1 Jan 2024 → 13 Jan 2024
- Accident data: 1 Jan 2024 → 27 Jul 2024

2.3 Data Type Conversion

The Date_Time columns were converted from string format to datetime format to support time-based analysis. After conversion, no missing values were introduced.

CHAPTER 3: PYTHON EDA

3.1 Descriptive Statistics

Understand the numerical characteristics of traffic and accident data.

SENSOR DATA – DESCRIPTIVE STATISTICS			ACCIDENT DATA – DESCRIPTIVE STATISTICS		
	Vehicle_Count	Average_Speed		Number_of_Vehicles	Casualties
count	300.000000	300.000000	count	5000.000000	5000.000000
mean	278.743333	49.100000	mean	2.518200	4.493400
std	125.767174	17.890694	std	1.107932	2.885069
min	52.000000	20.000000	min	1.000000	0.000000
25%	176.500000	33.000000	25%	2.000000	2.000000
50%	284.500000	48.000000	50%	3.000000	4.000000
75%	384.250000	64.000000	75%	4.000000	7.000000
max	494.000000	79.000000	max	4.000000	9.000000

3.2 Category Analysis

ACCIDENT DATA – ROAD CONDITION		SENSOR DATA – CONGESTION LEVEL	
Road_Condition		Congestion_Level	
Icy	1288	Low	104
Dry	1257	Moderate	100
Wet	1233	High	96
Under Construction	1222	Name: count, dtype: int64	
Name: count, dtype: int64		ACCIDENT DATA – ACCIDENT SEVERITY	
ACCIDENT DATA – TRAFFIC DENSITY		Accident_Severity	
Traffic_Density		Minor	1262
Moderate	1710	Fatal	1253
High	1687	Moderate	1248
Low	1603	Severe	1237
Name: count, dtype: int64		Name: count, dtype: int64	

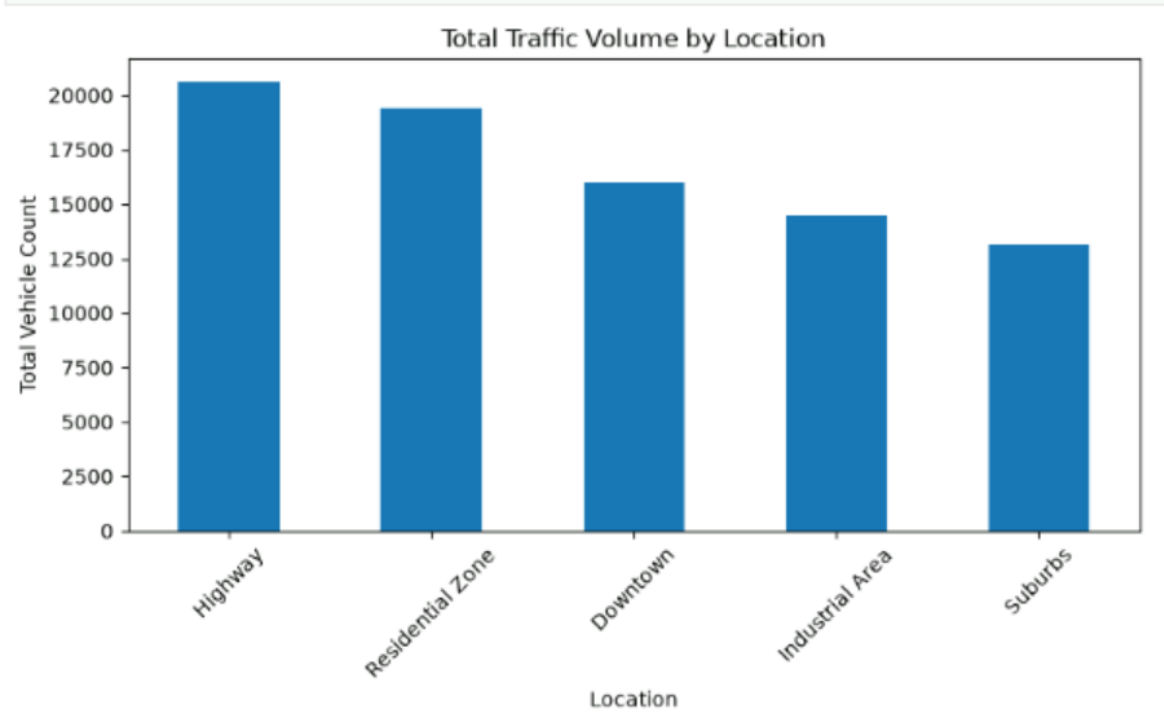
3.3 Location Analysis

ACCIDENTS BY LOCATION		
	Accident_Count	Total_Casualties
Location		
Highway	1018	4666
Suburbs	1017	4480
Residential Zone	998	4451
Industrial Area	993	4465
Downtown	974	4405

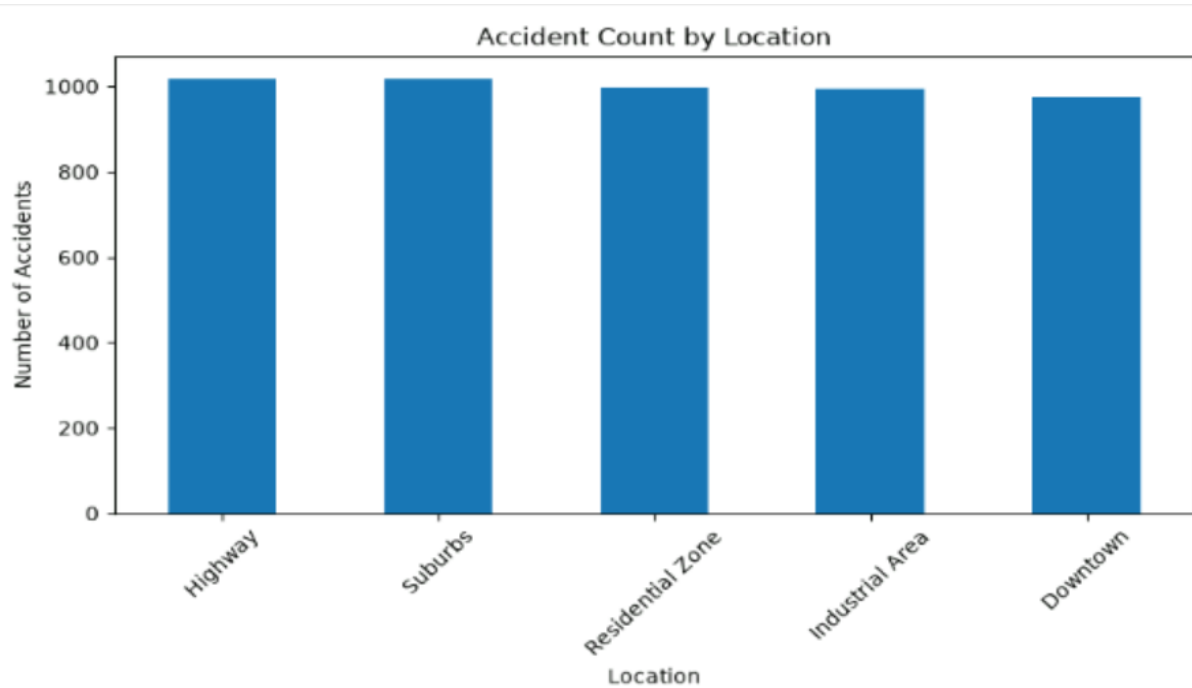
3.4 Time-Based Analysis

ACCIDENTS BY HOUR		
	Accident_Count	Total_Casualties
Hour		
0	209	940
1	209	937
2	209	898
3	209	970

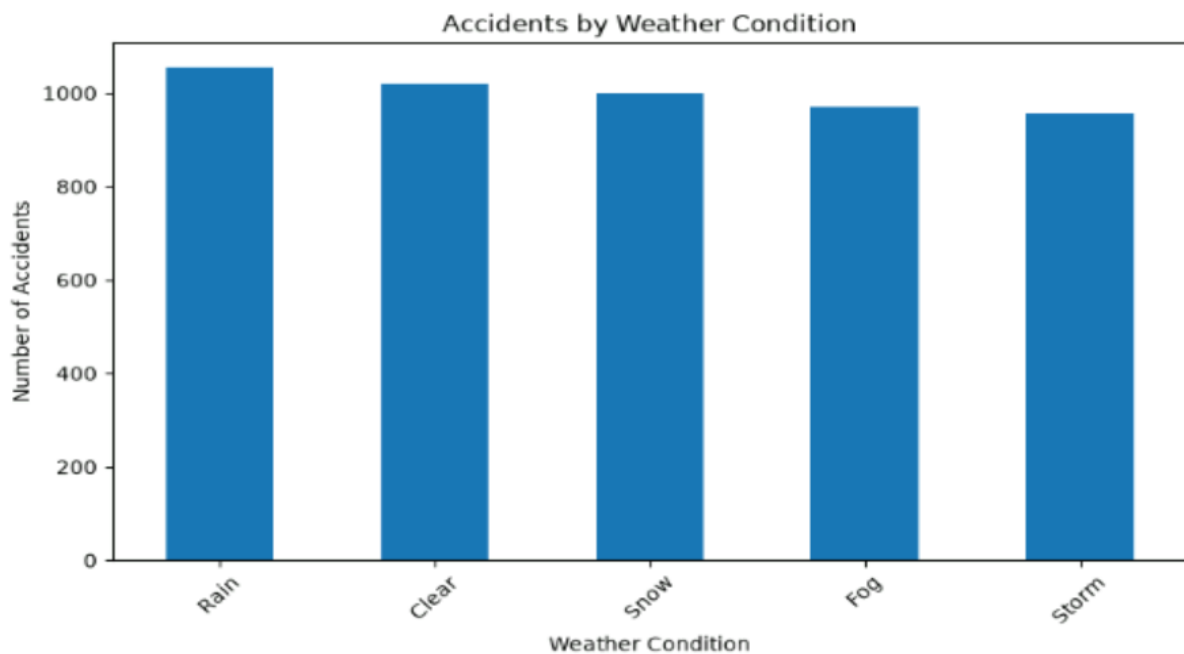
3.5 Analysis Charts



Highway recorded the highest total vehicle count with 20,623 vehicles, followed by Residential Zone with 19,436. Suburbs recorded the lowest total vehicle count with 13,110.



Highway recorded the highest number of accidents with 1,018 cases, followed closely by Suburbs with 1,017 cases. Downtown recorded the lowest number with 974 cases.



Rain was the most frequently recorded weather condition with 1,054 accidents. Icy road conditions were the most frequently recorded road condition with 1,288 accidents.

3.6 Initial Findings

The analysis shows that Highway recorded both the highest traffic volume and the highest accident count among the five locations. The accident data also shows relatively similar counts across severity categories, while Rain and Icy Road conditions were the most frequently observed environmental factors.

CHAPTER 4: MYSQL DATA MODELING AND SQL ANALYSIS

4.1 Create Database

Create Database and Import Data into Database/Datawarehouse. For SQL Analysis.

4.2 SQL Analysis

	Location	Total_Vehicles	Average_Speed
►	Highway	20623	49.71
	Residential Zone	19436	45.54
	Downtown	15981	49.36
	Industrial Area	14473	50.18
	Suburbs	13110	50.76

0.1:Traffic Analysis by Location

	Location	Accident_Count	Total_Casualties
►	Highway	1018	4666
	Suburbs	1017	4480
	Residential Zone	998	4451
	Industrial Area	993	4465
	Downtown	974	4405

0.2:Accident Severity Analysis

	Traffic_Density	Accident_Count	Total_Casualties
►	Moderate	1710	7701
	High	1687	7716
	Low	1603	7050

CHAPTER 5: POWER BI DATA MODEL AND DASHBOARD

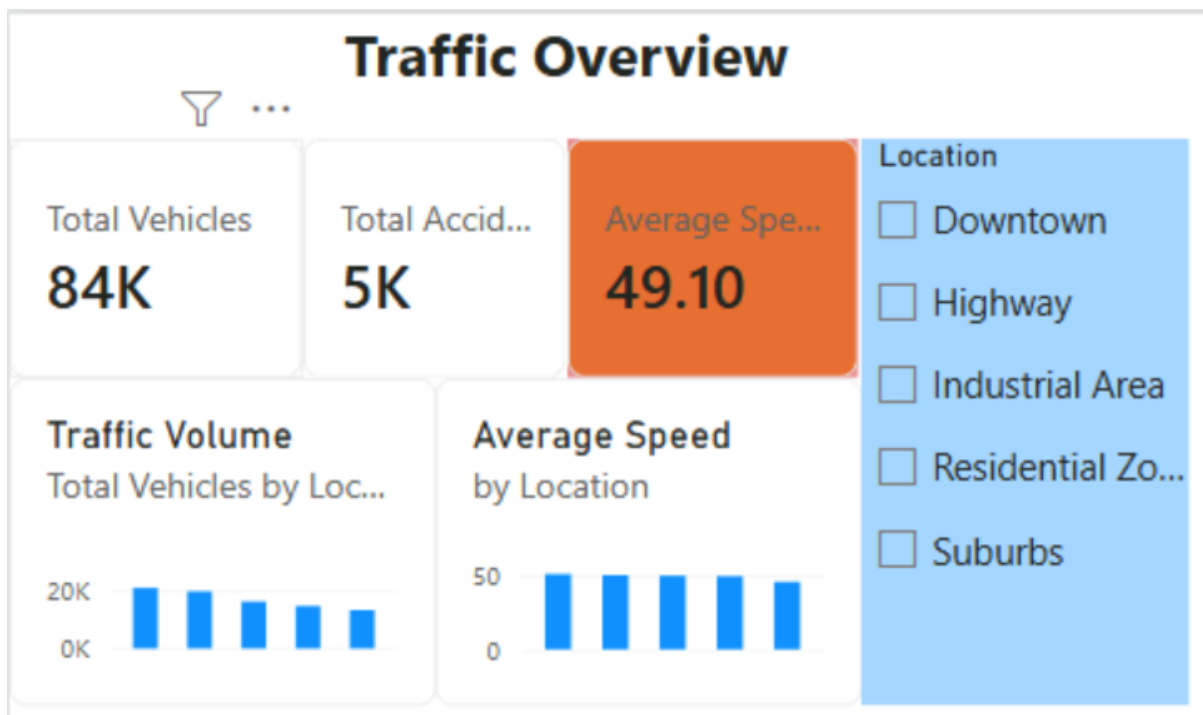
5.1 Power BI Data Import

Connected MySQL Server to Power BI.

5.2 Create DAX Measures

DAX measures were created to calculate important business metrics including total vehicles, average speed, total accidents, total casualties, average casualties, and severe/fatal accident counts.

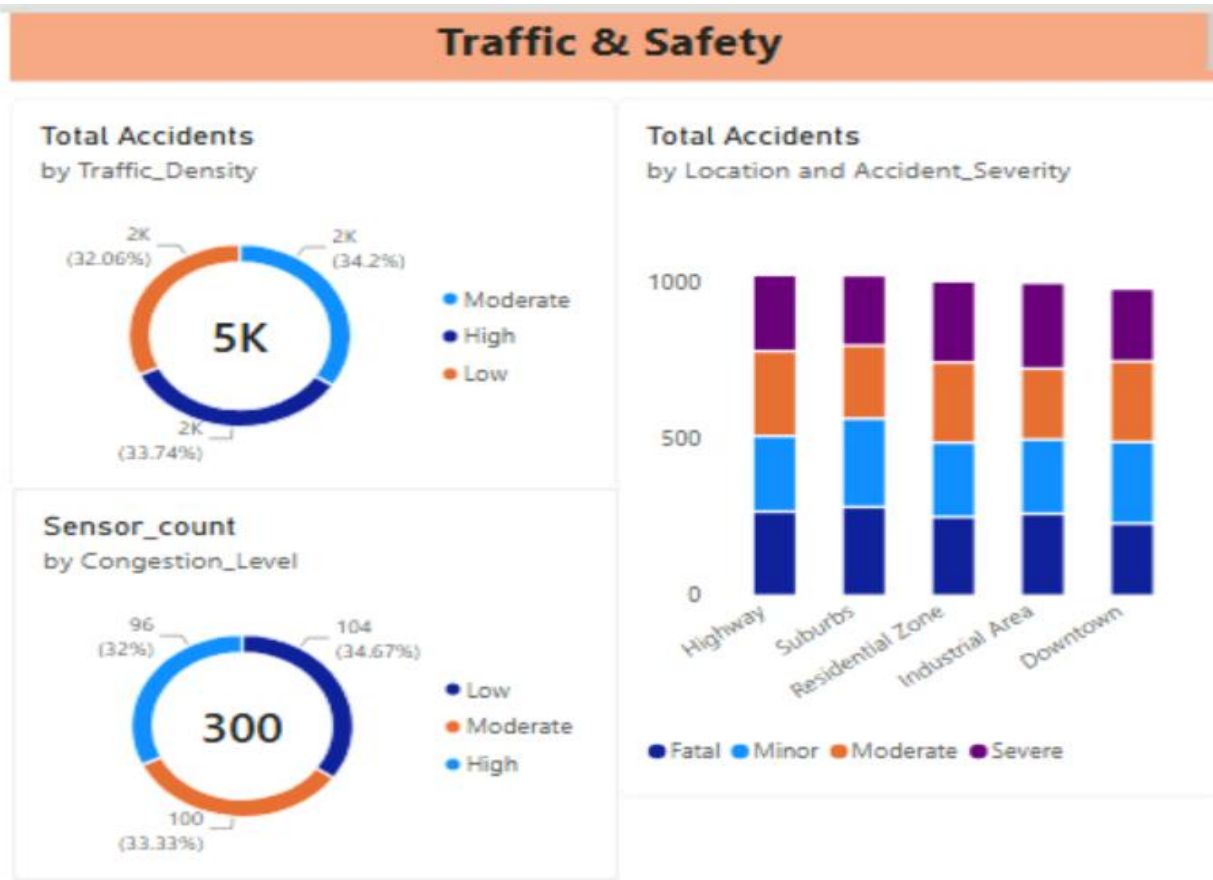
5.3 Dashboard Page 1: Traffic Overview



5.4 Dashboard Page 2: Accident Analysis



5.5 Dashboard Page 3 — Traffic & Safety



CHAPTER 6: RESULTS AND BUSINESS INSIGHTS

6.1 Traffic Findings

The traffic analysis showed an average vehicle count of 278.74 and an average speed of 49.10. Highway recorded the highest total traffic volume with 20,623 vehicles, followed by Residential Zone with 19,436 vehicles.

6.2 Accident Findings

Highway recorded the highest number of accidents with 1,018 cases, followed closely by Suburbs with 1,017 cases. Downtown recorded the lowest number with 974 cases.

6.3 Accident Severity Findings

Minor accidents were the most frequently recorded category with 1,262 cases, followed by Fatal accidents with 1,253 cases, Moderate accidents with 1,248 cases, and Severe accidents with 1,237 cases.

6.4 Environmental Findings

Rain was the most frequently recorded weather condition with 1,054 accident records. Icy road conditions were the most frequently recorded road condition with 1,288 accident records.

6.5 Traffic Density Findings

Moderate traffic density was the most frequently recorded category with 1,710 accident records, followed by High traffic density with 1,687 and Low traffic density with 1,603.

6.6 Business Insights

The analysis provides a consolidated view of traffic activity and accident patterns across the five locations. Highway requires particular attention because it recorded the highest traffic volume and the highest accident count in the analyzed data.